Drift Sense, Phase 2 failure analysis

Team Atlas. Every number is traceable to a run under `experiments/` in the repository, github.com/kushal-script/Atlas; nothing is estimated.

Three limits cap this pipeline, and none of them is an optimisation failure. A single pitch lattice is ambiguous in the pixel domain, so the evidence itself prefers the wrong site. The correlation surface carries a fixed amount of information about presence, and every statistic derived from its peak is a rescaling of one already used. The sensor's eight bit tone map removes the variance structure that the standard statistical correction for shot noise exists to restore. Each was measured rather than asserted, and each bounds a different scored component.

The structural limit, and the oracle that bounds it

The dominant failure is the impostor that rescales onto the lattice. An absent pair's reference comes from a different die region of the same architecture. Because the zoom is unknown over 8 to 12, the pose search is free to rescale the impostor until its lattice period locks onto the search image's lattice, and the shared periodic content then correlates respectably. Measured over a 216 pair generated suite, absent pairs reach a median peak of 0.56, higher than genuinely present pairs at severity 2 degradation (0.52). A threshold on the correlation peak alone therefore cannot implement the found flag: its best reject class F1 on that suite is 0.46, with the degenerate never reject rule close behind, which is exactly the failure the addendum warns scores zero.

The limit is bounded, not estimated. An oracle probe hands the matcher the true zoom and rotation from the generator's own record, which removes the pose search from the problem entirely. The true site still wins the correlation on only 88 percent of severity 1 pairs, 84 at severity 2, 64 at severity 3 and 34 at severity 4, a monotonic decline across the whole ladder rather than a cliff at one rung, and at the hardest level the winning impostor leads by a median of 0.054, nearly three times the 0.02 margin an alternative must beat to overturn the answer. A perfect pose search would therefore localize about one severity 4 pair in three; the shipped pipeline scores about one in five. That gap is not reachable by any better search, since the correlation already prefers the wrong site, but it is reachable by asking a different question of the same pixels, and an earlier version of this document overreached in saying otherwise. Normalized cross correlation divides each window's mismatch by that window's own contrast, so a high contrast impostor is forgiven exactly the mismatch that should convict it, and on a degraded capture whose true site has lost contrast with the dose that forgiveness decides the match. The absolute residual after a gain and offset fit forgives nothing, and ranked over the same candidate peaks it prefers the true site on 43 percent of held out severity 4 pairs against correlation's 14, and 50 against 8 on the training suite. Gated on a relative margin of 0.045 fitted on the training suite alone and fixed before any held out number was read, it is worth plus 0.94 of 40 end to end on the held out suite and plus 0.47 on the training suite, the whole effect inside the degraded set with the nominal set unmoved to three decimals and runtime unchanged (`experiments/20260901_residual_rms_rerank`). What remains information limited is the 65 percent of the severity 4 loss belonging to dose and beam spot, which is signal never collected; the rest was the limit of one evidence function rather than of the data. Held out credit is 0.667 on the degraded set against 0.786 nominal, falling monotonically 0.788, 0.637, 0.433, 0.206 across a severity balanced suite.

Rotation is reported to a convention, not an assumption. After a final euclidean alignment polish the rotation credit is 0.881 on held out pairs; the motion model is deliberately euclidean because an affine fit on a periodic lattice slides scale by pitch fractions and made the scale estimate worse. The reported sign is a single documented constant, settled against the organiser sample pairs rather than assumed from our own generator.

The severe set's loss, attributed factor by factor. Calling a limit physical is a claim, so it was measured. The severity ladder pushes six factors at once, which leaves no attribution, so paired suites were generated from identical seeds, a control under the full severity 4 ladder and one suite per factor with that factor alone restored to nominal, localized without the presence decision so the difference belongs to the factor rather than to a threshold. Against a control credit of 0.200, restoring electron dose alone returns 0.550 and drops the median error from 500 px to 1.22 px; the beam spot returns 0.370, scan jitter 0.310, polygon scaling 0.270, while drift and charging return 0.170 and 0.150, nothing outside the noise of twenty pairs, and charging could not have moved on that design at all, its twelve DRAM pairs carrying no material the mask keyed on, a generator no op since corrected.

Dose and beam spot are 43.8 and 21.2 percent of the attributable loss, jitter and polygon scaling 13.7 and 8.8, so **65 percent of the whole severity 4 loss belongs to shot noise and optical blur and 22 percent to structured corruption**, the remaining 12 or so percent being interaction the single factor design cannot attribute; measured against the attributable part alone the split is 74 to 26. The two halves differ in kind. Shot noise is information never collected: a dose of fifteen percent of nominal means the electrons that would have carried the missing spatial frequencies were never emitted, and no filter, no deconvolution and no learned prior recovers a measurement that was not made. Beam spot blur is the same statement in the frequency domain, a physical low pass whose stopband holds no signal to restore. Jitter is the opposite kind of damage, a structured displacement of rows that in principle inverts, and it was priced rather than assumed: a row shift estimator that inverted it perfectly would lift severity 4 credit from 0.200 to at most 0.310, and severity 4 is about a fifth of the degraded set, so the ceiling is half a localization point, before accounting for the fact that the estimator would have to measure row displacements from the same shot noise that dominates the loss. It was not built.

The information limit on the found flag

Presence is decided by combining the peak with the evidence that does not rescale: how degenerate the correlation surface is, whether the wide pose search beat the nominal pose, whether the deviation field, what remains after the shared periodic content is projected out, actually matched anywhere, and, since the September refit, whether a second evidence function agrees. A fitted battery of seven raw pixel statistics re ranks the top correlation peaks, and the decision reads that battery's probability for the chosen site, its margin over the runner up, and its agreement with the correlation's choice; disagreement between two independent evidence functions over which site is true is close to a direct measurement of ambiguity. The rejection F1 is scored over all 180 grayscale pairs, of which 40 carry no true instance. On the held out proportional holdout the combined decision reaches a reject class F1 of 0.800 against 0.46 for the best single feature and 0.655 for the fifteen surface diagnostics alone, at an operating threshold set by sweeping every candidate against localization, pose, rejection and calibration together over pooled suites; an earlier sweep caught the fitted threshold refusing a third of a batch whose disclosed composition is 22 percent absent, and a batch adaptive rule built to survive a calibration shift was simulated under logistic shifts and rejected, since the strongly bimodal probabilities barely move across any fixed boundary and the rule cost more unshifted than it recovered shifted. The operating point itself moved once more when the organisers' released reference scorer settled which class the graded F1 is computed on: its shipped confusion arithmetic, and the addendum's own definitions of the false positive as a found absent and the false negative as a rejected present, both grade the found class, where this pipeline had optimised the reject class. The threshold ships at 0.35, chosen on a five suite pooled sweep of 492 pairs over both generators and the hardest decoys; on the four suite pool it keeps three quarters of the found graded reading's gain, is positive on every suite under it, and costs 0.16 pooled under the reject class reading, inside run noise. Under the found class reading their scorer implements it reads 0.94 to 0.97, above the 0.90 gate; the ambiguity between the two readings is priced by that hedge rather than argued away.

Why the ceiling was where it was, and what moved it. Sweeping every threshold against the blind sized holdout once put the reachable reject F1 at 0.667, and closing the distance to the 0.90 bonus gate needed signal that is not on the correlation surface. Exhaustive feature work established that it is not there. A teammate branch carrying a 25 feature set was ablated over identical records and folds: its ten additions to our fifteen contributed 0.0075 in total, with three peak ratio features actively harmful and six others exactly neutral. Peak to correlation energy contributed nothing, 0.7040 with and 0.7040 without. An extreme value model was the most promising candidate and the most instructive failure: fitting a Gumbel to the correlation surface with the peak neighbourhood excluded separates present from absent at 1.02 standard deviations, twice the 0.52 of the prominence feature already shipped, yet it correlates at $r = 0.856$ with that feature's logarithm and contributes minus 0.0003 cross validated. Three candidates failed in exactly this way, which was enough to state as a rule: any statistic derived from the height, shape or tail of the correlation peak is a monotone rescaling of one already present, and the features that carry independent signal come from a different construction, such as quadrant agreement, which separates at 3.0 standard deviations and splits the classes at 77 percent on its own. The September refit is the rule's confirmation from the constructive side. The re rank battery reads raw pixel residuals rather than the correlation surface, and its disagreement features lifted the reject F1 on the proportional holdout from 0.655 to 0.800 and on the blind sized suite from 0.557 to 0.605, gains a surface derived statistic could not have carried; the three candidates built afterwards from the surface again, a period aware second peak ratio, the peak curvature and the reference's spectral balance, were measured and declined, winning on one suite of three. The 0.667 figure was the reachable F1 of one evidence function's ranking on one suite, not a property of the problem; each added evidence function sets its own ceiling. The residual weakness is located and taken to its boundary: identical geometry decoys are grabbed at about six in ten, a cost priced into the pooled threshold selection. Their discriminating signal was found, the winning site's absolute deviation correlation at d prime 1.68 on decoys, but it is absent from the own fitting domain at 0.02 and all five measured deployments damage a held out suite, so the weakness stays priced and the signal recorded (`experiments/20260902_presence_dev_refit`). A third evidence function then arrived from the organisers' released generator, which regenerates every present pair until the raw full reference correlation's global argmax sits on the label with a margin of at least 0.02, a statistic this pipeline had never computed because the bandpass removes the zone scale content it reads. Computing it once at the estimated pose and letting it override the answer when it disagrees and clears a margin floor was swept on the fitting suites and judged held out at plus 1.29, plus 0.86 and exactly zero of 40, and on the organisers' untouched sample it changed exactly one row, the severity 4 pair, from a 426 px miss to 0.31 px. The floor was then hardened after an external audit reproduced the override's one destructive mode, a wrong lattice grab at margin 0.0202: fresh override off records over six suites put every observed damage at 0.0202 or below and every genuine raw resolution at 0.1089 or above, the sample's severity 4 pair at 0.1453, so the floor moved from 0.02 to 0.05, clearing the damage band by 2.5x, keeping every rescue, and moving every held out suite by exactly zero (`experiments/20260902_raw_override_hardening`).

The low precision is not the loss it appears to be. Over 150 degraded pairs the decision rejects 52 present ones and only 2 of those would have earned any credit, the rest missing by a median of 458 px, so the presence decision doubles as self error detection. Rejecting nothing at all would score 0.535 against the 0.521 it scores now, placing the decision within 0.014 of optimal and ruling out a looser threshold as a source of points.

A shortcut our own generator almost taught us. The first fitted presence model put its largest weight on the measured image noise, because in the first training suite every absent pair happened to be a clean capture, so noisy meant present. That rule inverts on any blind set whose absent pairs are degraded. The generator now degrades half its absent pairs, yet the shipped model keeps its largest weight on noise: the refit on the degraded absent suite was measured and declined, losing seven of eight judge suites including the organiser recipe ones, so at this operating point noise is presence evidence, not a shortcut. Relatedly, the Phase 1 residual stage decided ties with a z score, whether one candidate stands out among its peers; an impostor site can stand out among other impostor sites by chance, and measured medians confirmed it separates nothing, 3.3 present against 3.4 absent. The shipped feature is the winning site's deviation field margin in robust units, which asks whether the site matched at all rather than whether it beat its neighbours.

The sensor limit, and a correction that cannot be applied

Electron arrival is Poisson, so variance tracks the mean, while normalized cross correlation is optimal for constant variance noise. The standard remedy is the generalised Anscombe transform, which stabilises the variance before correlating, and elevated shot noise is the single largest attributable factor in the severe set, so the transform should have been worth points. It is not, because the data reaching the matcher is not raw electron counts.

Local variance was measured over 56250 flat eight by eight tiles from twelve search captures, binned by local mean and keeping only the flattest thirty percent so the quantity measured is noise rather than structure. Under a Poisson law the ratio of variance to mean is constant. Measured, it falls monotonically from 16.3 in the darkest bin to 7.1 in the brightest, while the raw variance moves from 721.6 to 1125.0 and is nearly flat across the middle of the range; comparing the darker half of the range against the brighter gives 1043 against 1143, a ratio of 1.10 across an eightfold change in brightness. The eight bit tone map applied during capture has already equalised most of the variance the transform exists to equalise, so the correction has little left to correct and can only bend the relation the other way. This is reported as a premise that fails rather than a transform that harms: an end to end trial was run, but with every feature rescaled the presence model, fitted on untransformed inputs, rejected the entire suite, so that trial measures a stale model and not the transform. The honest statement is that the transform is answering a question this data no longer asks, and that recovering it would require the raw counts the pipeline does not emit.

What was measured and declined

Width rescue helps, but not where it was aimed. Polygon scaling changes feature widths between the captures, so re matching under a small morphological width change, the imaging analogue of a CD offset, is a natural counter, applied when the first pass shows the mislock signature of a weak peak on a degenerate surface. Held out measurement placed the gain precisely, and not where the design intended: on the twenty severe present pairs it changed nothing at all, identical credit of 0.160 and the same four pairs inside five pixels, while nominal credit rose from 0.690 to 0.738. The severe failures are not width mismatch but the same ambiguity the oracle characterised. Two width variants capture the entire effect; four returned an identical core score for 1.3 seconds more per pair. An earlier probe suggested more only because it selected among variants using the true error, an oracle no scored run has.

The optical bonus set was blocked by our own physics. Rendering the optical reference through a literal Abbe limited point spread, 0.21 lambda over the numerical aperture, is about 126 nm wide, which at one nanometre per pixel is a 126 pixel blur across a 1000 pixel frame. The reference came out a smooth colour gradient while the wide image kept its structure, so the high magnification image held less information than the wide one and the registration task did not exist; credit sat at 0.10 against a gate of 0.40. The physics is right, which is why no real optical tool images a one micron field, but the addendum asks for an analogue with the reference present and scores it, so the analogue now samples the point spread relative to the pixel as the electron model does, keeping the three interference coloured channels, photon limited noise, vignetting, tilt and wavelength scaled sharpness. Credit rose from that one change to 0.444, 0.633, 0.619 and 0.430 on four suites under disjoint seeds, pooling to 0.529 over 192 pairs. Bootstrapping 20 pair draws puts the fifth percentile at 0.35 and the chance of missing the gate near 11.6 percent; an earlier estimate over the first three suites read near two percent, and a fourth suite returning 0.430 on 60 pairs raised it almost sixfold, which is the evidence that between suite spread dominates the within suite draw and that a risk read off any single suite understates it; the twelve organiser recipe optical pairs all read 1.000, so the risk assumes blind optics resembling ours rather than the recipe's. Summing the chroma plane surfaces scored 0.24 against 0.42 at twice the runtime. Widening the blur bank was equally unproductive: 0.411, 0.406 and 0.394 against the shipped 0.433, so the entire optical gain came from correcting the generator rather than from tuning the matcher.

The defects self scoring cannot find

An adversarial audit of our own entry point, six independent reviews with every finding put to a separate reviewer instructed to refute it, returned seventeen surviving findings. The two that mattered most cost nothing on any suite we own, because they fire only on input shapes our generator never produces. A pairs csv saved from a spreadsheet carries a byte order mark that stays glued to the first header name and survives strip, since it is not whitespace; the pair_id column then goes unrecognised, every row is named row_0000 onward, and the run exits cleanly having written a well formed file in which not one row joins to the truth. A grayscale capture exported as a three or four channel png was routed to the optical preset because the modality was decided from the array's rank rather than from whether the planes differ; on identical pixels that halved credit. A later pass found a third of the same kind: one data row carrying a surplus field puts a null key into the parsed row, and the exception escaped the pair loop before the per pair handler could see it, so a single stray comma produced no predictions file at all and forfeited every pair rather than the one that was malformed. All three are small fixes, all were reproduced before and after, and the entry point is now verified against a byte order mark, a replicated colour channel, ragged and short rows, duplicate ids and a missing image, with the packaging self test running it under an audit hook that names any file opened outside the supplied paths. A suite you generate cannot exercise the ways a file you did not write can be shaped.

Two further defects were real and cost less than expected. The contrast polarity decision rebuilt the full resolution correlator but not the prescreen one, so on the two pairs in five where inversion fires the wide grid was ranked by the negated correlation; correcting it moved the core score by two hundredths of a point, because the full resolution stage recovers what the prescreen loses. It ships corrected anyway, since a knowingly inverted ranking is a defect whether or not this generator punishes it. The optical preset also asked for a 1200 px Gaussian on a 1000 px frame, computing the image mean convolutionally, and now returns it in closed form for identical credit.

A measurement hazard worth recording. The wall clock guard that protects against the scored timeout also makes answers depend on machine load, because a pair that runs slow has its optional stages skipped before they start. Sweeping eight configurations in parallel produced a table in which the shipped configuration scored 0.344, while the same configuration on the same pairs run alone scored 0.433, a relative difference of a quarter from contention alone. Run one at a time the guard never fires: raising its budget from nine to fifteen seconds changed no answer and no credit. Configuration comparisons for this system have to run one at a time, and every number reported here does. The same discipline applies to the interpreter: the reference machine runs Python 3.11, predictions are byte for byte identical to the development interpreter and credit is unchanged, but the reference stack is 12 percent slower at the median, so accuracy measured during development transfers and runtime does not.

The posture on rejection

False positives and false negatives are not symmetric on a tool: a false grab silently corrupts a measurement, a false reject costs one cheap rescan. Where thresholds tie on F1 this submission takes the more precise one, and the score column expresses confidence in the decision made, so a rejection on overwhelming evidence ranks as high as a confident detection.